Vinothini Balasubramani

Data Analyst
Chennai, India (Relocating to Dublin) |Email: vinothinibs09@gmail.com | LinkedIn: linkedin.com/in/vinothini-balasubramani
GitHub: github.com/VinothiniBS | Portfolio: vinothinibs.github.io/portfolio

Eligible for Critical Skills Employment Permit

PROFESSIONAL SUMMARY

Data Analyst with 6+ years of experience, an MSc from University College Dublin, and prior onsite experience in Dublin. My background spans software engineering, data integration, and analytics including predictive modelling and forecasting, giving me a broad perspective that helps me build solutions others can trust and act on. I am happiest when the work I do makes someone else's job meaningfully easier.

CORE SKILLS

Analytics & BI: Power BI, Looker Studio, Tableau, Advanced Excel, KPI Dashboarding

Programming: Python (Pandas, NumPy, Flask), SQL, R, JavaScript

Data Engineering & Cloud: ETL, Google Cloud Platform, BigQuery, REST API Integration, SAP CPI/PO, Oracle, MySQL

Techniques: Data Cleaning, Predictive Modelling, Forecasting, Customer Segmentation (RFM), Automated Reporting, Data Quality Validation, Anomaly Detection, Machine Learning

Tools & Platforms: Git, Google Apps Script, Jupyter, dbt, Airflow, Zapier, Docker

EMPLOYMENT HISTORY

Analytics and Automation Engineer

Assay Genie, Remote| Jan 2026 - Present

- Developed a **Python-based** application to automate bulk product datasheet generation across all product categories, replacing manual workflows and accelerating the introduction of 50+ new products per month.
- Migrated and scaled the QC application to **GCP and BigQuery** for deep-level data sheet validation across 2/3 of product categories and integrated the **Claude API** for a specific category to generate product manuals from predefined templates, improving accuracy and performance over the previous system.
- Maintain and enhance **BigCommerce and Doofinder** data pipelines using Python and Google Apps Script across 120,000+ product records, including an automation that syncs citations data from Google Sheets to the matching product pages and refreshes the citations landing page live on the website, with an automated weekly citations report.
- Integrated price file data into a **Cloud SQL database** on GCP to centralise pricing in a single queryable source and now handle bulk price updates and corrections through SQL queries, alongside routine data quality checks on the pricing data using Python; maintain and update the internal UI used to manage it.
- Enhance and maintain Data Studio dashboards and generate **operational reports** across **logistics, sales, and leadership** teams, building on the self-service analytics foundation established in the previous role.

Data Analyst

Assay Genie, Dublin, Ireland (Onsite) | Jan 2024 - Mar 2025

- Designed **Data Studio dashboards** for logistics and sales teams, integrating **Cin7 API** data via Python and Google Apps Script to automate weekly performance reporting and deliver real-time operational tracking, saving **8+ hours per month**.
- Developed an automated **QC and datasheet generation** application covering **1/3 of product categories**, reducing documentation preparation time and enabling faster quality checks for the team.
- Built **Python-based predictive models** that improved inventory accuracy and reduced overstock by a quarter of usual levels, directly impacting working capital.
- Maintained and managed a **price file of 100,000+ products** in Google Sheets, handling product additions, modifications, and data accuracy across the full catalogue.

Data Integration Engineer
Tech Mahindra, Chennai, India | Apr 2020 - Jun 2022

- Designed and delivered **50+ SAP CPI/PO integration** pipelines for enterprise clients, **reducing sync errors by 40%** and ensuring reliable data flow across mission-critical business systems.
- Resolved **100+ EDI transformation issues** with consistent turnaround, streamlining data workflows and improving resolution speed by 35%, ensuring reliable data availability for clients.
- Developed SQL scripts for database schema design and data loading across Oracle databases, supporting accurate and consistent data availability for downstream reporting teams.

Full Stack Developer
Tech Mahindra, Chennai, India | Jun 2018 - Mar 2020

- Developed **responsive Angular UI components** for a major telecommunications client, delivering a polished internal application experience used across the organisation.
- Enhanced **Java and MySQL** backend processes for a banking client, reducing query execution time by 20% and improving the reliability of core financial workflows.
- Partnered with QA and DevOps teams to manage feature rollouts end to end, achieving a **98% deployment success rate** across all releases.

EDUCATION

MSc in Data and Computational Science, Sep 2022 - Aug 2023
University College Dublin, Ireland | Second Class Honours, Grade 1 (2H1)
Bachelor of Engineering in Mechatronics, Aug 2013 - Apr 2017
SNS Institutions, India | First Class with Distinction (8.3/10)

CERTIFICATIONS

Career Essentials in Data Analytics - Microsoft & LinkedIn | Dec 2025
MySQL for Data Analytics and BI - Udemy | Dec 2025

PROJECTS

Dublin Bikes Data Pipeline (Python, Airflow, dbt, BigQuery, Docker)
- Built an end-to-end pipeline ingesting live Dublin Bikes data every 15 minutes via the JCDecaux API into BigQuery, transformed with dbt into tested staging and marts layers (14 data-quality tests), orchestrated with Airflow in Docker.
- Surfaced station-health insights through a Data Studio dashboard, identifying north-city stations empty over 50% of the time as a rebalancing signal.

E-Commerce Sales Analysis & Customer Segmentation (Python, Prophet, RFM)
- Analysed e-commerce transaction data to surface monthly trends, top products, and country-level demand, built a Prophet model forecasting sales for 30 days ahead.
- Implemented RFM (Recency, Frequency, Monetary) customer segmentation to enable targeted marketing and retention strategies.

COVID-19 Vaccination Impact Analysis (R)
- Built regression and anomaly-detection models to quantify the relationship between vaccination coverage and infection trends, supporting risk-focused public health assessment. (MSE: 0.0702)
- Developed stakeholder-ready visual reports to communicate trend deviations and policy impact insights.

Irish Weather Insights Dashboard (R Shiny)
- Designed an interactive R Shiny dashboard to compare climate trends across Dublin, Belmullet, and Malin, integrating historical metrics such as temperature, humidity, wind speed, and precipitation.
- Enabled city-level comparison and seasonal trend forecasting to support tourism and environmental monitoring.